Brett Longworth

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RESEARCH INTERESTS

- Dissolved gasses as tracers of ecosystem function in seawater.
- Microcontroller systems, sensor platforms, telemetry, and shipboard data integration.
- Field-deployed quadrupole mass spectrometry; radiocarbon as a tracer of carbon source.

SKILLS

- Carpentries-certified instructor, course organization, lesson development
- R, Python, C++/MicroPython (microcontrollers), Shell, git/GitHub, and data-driven web applications.
- Data pipelines, statistics for data reduction and QA/QC, cloud and local deployment, agentic coding.
- Shipboard and laboratory mass spectrometry, chemical analysis, sensor platform and telemetry development.

PROFESSIONAL EXPERIENCE

Mar 2026 – Present Woods Hole Oceanographic Institution — *OOI*
Sensor tank testing, discrete pH/DIC sampling and analysis; data reduction package for T, S, pH and DIC.

Jan 2023 – Present Woods Hole Oceanographic Institution — *Machine Lab*
Microcontroller code, telemetry, and Gb–Tb data pipelines for mass spectrometer and ADV sensor platforms.

Nov 2021 – Present Woods Hole Oceanographic Institution — *NES-LTER Cruises*
O2/Ar by equilibrator inlet mass spectrometry, dissolved gas sampling, CTD/rosette deck and lab operations.

Nov 2023 – Jan 2026 Woods Hole Oceanographic Institution — *LOC-NESS Project*
Real-time web mapping, shipboard sensor interfaces, CTD/drifter operations, and QA/QC of underway data.

Mar 2006 – Nov 2023 Woods Hole Oceanographic Institution — *NOSAMS Research Associate*
Operation, maintenance, and development of radiocarbon accelerator mass spectrometry systems.

2008 – 2013 Woods Hole Oceanographic Institution — *Research Cruises*
GEOTRACES EPZT (2013), GEOTRACES NAT (2011), ANDREX JC030 (2009), CLIVAR I6S (2008).

EDUCATION

2001 – 2005 University of Massachusetts, Amherst, MA
M.S. Geosciences, August 2005. Thesis: *Role of non-modern carbon pools in riverine and global carbon cycles.*

1995 – 1998 University of Massachusetts, Amherst, MA
B.S. Biology, concentration in ecology and evolution. Year exchange in Freiburg, Germany.

TEACHING, SERVICE & AWARDS

- **Teaching:** Organized and taught workshops in R, Python, git, shell, and SQL (2018 – present).
- **Service:** Ocean Informatics Working Group, Staff Mentoring, Technical Ladder Review Implementation Committee.
- **Awards:** Technical Staff Training Awards (2019, 2020, 2021); APO/DDVPSE funding for coding workshops (2023 – 2026).

SELECTED PUBLICATIONS

1. **Longworth, B. E.** *et al.* Radiocarbon measurement of CO2 from carbonate minerals using a hybrid gas ion source with an open split interface. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* **531,** 49–55 (2022).
2. **Longworth, B. E.**, Reden, K. F. von, Long, P. & Roberts, M. L. A high output, large acceptance injector for the NOSAMS Tandetron AMS system. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* **361,** 211–216 (2015).
3. **Longworth, B. E.** *et al.* Carbonate as sputter target material for rapid 14C AMS. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* **294,** 328–334 (2013).
4. **Longworth, B. E.**, Petsch, S. T., Raymond, P. A. & Bauer, J. E. Linking lithology and land use to sources of dissolved and particulate organic matter in headwaters of a temperate, passive-margin river system. *Geochimica et Cosmochimica Acta* **71,** 4233–4250 (2007).

Full publication list: orcid.org/0000-0001-8318-2054